

Calvin Hou

Seattle, WA | 206-601-6421 | hou61@gatech.edu | calvinhou.com | linkedin.com/in/houcalvin

Education

Georgia Institute of Technology

B.S. in Electrical Engineering

Concentration: Circuit Technology & Signal Processing

Coursework: Digital System Design, Programming for Hardware/Software Systems, Signal Processing, Circuits & Electronics

Atlanta, GA

May 2028 (Expected)

GPA: 3.9

Work Experience

Firmware Research and Design Intern

Delta Electronics

Bangkok, Thailand

June 2026 - August 2026

- Diagnosed EMF-induced signal faults in production data center PSUs via root-cause analysis and board-level debugging; implemented and validated 21 filter configurations (IIR averaging + configurable glitch filtering) in STM32 embedded C, cutting erroneous fault events and exposing tunable parameters as user-configurable endpoints.
- Identified fragmented, product-siloed MCAL implementations across Delta's CDBU power supply lines; refactored and standardized 6 STM32 embedded C modules (PWM, ADC, DAC, CAN FD, UART, Watchdog) — eliminating redundant rewrites for 8 engineers across 6 bespoke PSU/supercap designs serving Fortune 10 hyperscale cloud and semiconductor clients. First module merged and distributed team-wide.
- Authored exhaustive Python bench test suites per module: the PWM suite alone validated 900 hardware configurations (9 timers x 4 channels x 5 frequencies x 5 duty cycles), establishing min/max operational bounds and generating UL60730-1 certification evidence.

Advanced Robotics Centre Research Intern

National University of Singapore

Singapore

Sep 2024 - Oct 2024

- Wired and integrated robotic actuators for a 3-DOF robotic arm; designed and built a joystick/button controller; programmed an inverse-kinematics algorithm resolving joint angles for precise XYZ positioning — research focused on automating port lashing (cargo tie-down) operations in Singapore.

Hardware Research and Design Intern

Asia Vital Components

New Taipei City, Taiwan

Aug 2024 - Sep 2024

- Designed and validated liquid-to-air cooling assemblies for AI data center hardware, achieving a 30% thermal performance gain via K-type thermocouple/DAQ-based heat dissipation characterization that directly set production specs.

Projects

1S USB-C PD Battery Charger

Jan 2026 - Present

- Designed a 4-channel, 15W 1S Li-ion charger PCB in Altium (schematic capture through layout) with thermal and EMI mitigation; wrote ESP32 C++ firmware to show real-time SOC/thermal monitoring over I²C to an OLED display

Engineering Projects & Teams

Avionics Team Member

YJSP (Yellow Jacket Space Program)

Atlanta, GA

Jan 2026 – Present

- Designed STM32-based PCB as Responsible Engineer on the Control Test Vehicle, controlling 2 linear actuators, 4 RCS fans, and 2 brushless ESCs via PWM; took the board from schematic through layout in Altium and sent it to fab.
- Designed full ADIN1200 Ethernet PHY circuit and PCB layout for signal integrity in a high-EMI flight environment; designed multi-rail power system (22.2V to 12V/4.4V/3.3V) with STM32 ADC-based telemetry on each rail.

Electrical Systems Member

Hytech Racing (Electric Formula at Georgia Tech)

Atlanta, GA

Aug 2025 – Present

- Led full-lifecycle integration of 6 DS18B20 temperature sensors across battery pack modules, achieving FSAE EM.3.3 compliance through schematic design, 1-Wire firmware, EMI-compliant placement, and on-vehicle validation.
- Introduced J1772 inlet to accumulator charging system, enabling 240V access at competitions and in workspace — reducing charge time by 50%.

Team Founder, President, Mechanical and Electrical Lead

FIRST Robotics Team 9442

Seattle, WA

Jun 2023 – Apr 2025

- Founded and led a 15-member FRC team as sole technical lead — raised \$45,000+; designed and integrated electrical/control systems (CAN motor control, power distribution, sensor integration) across 4 competition robots, achieving 0 electrical failures across 10+ competitions and a top 4% global ranking among 3,500+ teams.

Skills & Awards

PCB & Circuit Design: Schematic Design & PCB Layout (Altium, KiCad) · Power Electronics · Signal Integrity · Board-Level Debug · Digital Design · Root Cause Analysis

Test & Lab Equipment: Oscilloscope · Logic Analyzer · Multimeter · DAQ Systems · Soldering · Wire Harnessing & Crimping

Embedded Systems: STM32 Embedded C · C/C++ · Python (Instrument Control, Test Automation)

Languages: English (Native), Chinese (Professional)

Awards: 1st in Technical Skill, Amazon Future Engineer · District Event Winner (FRC) · 1st in State, DECA VBC · TSA Semifinalist